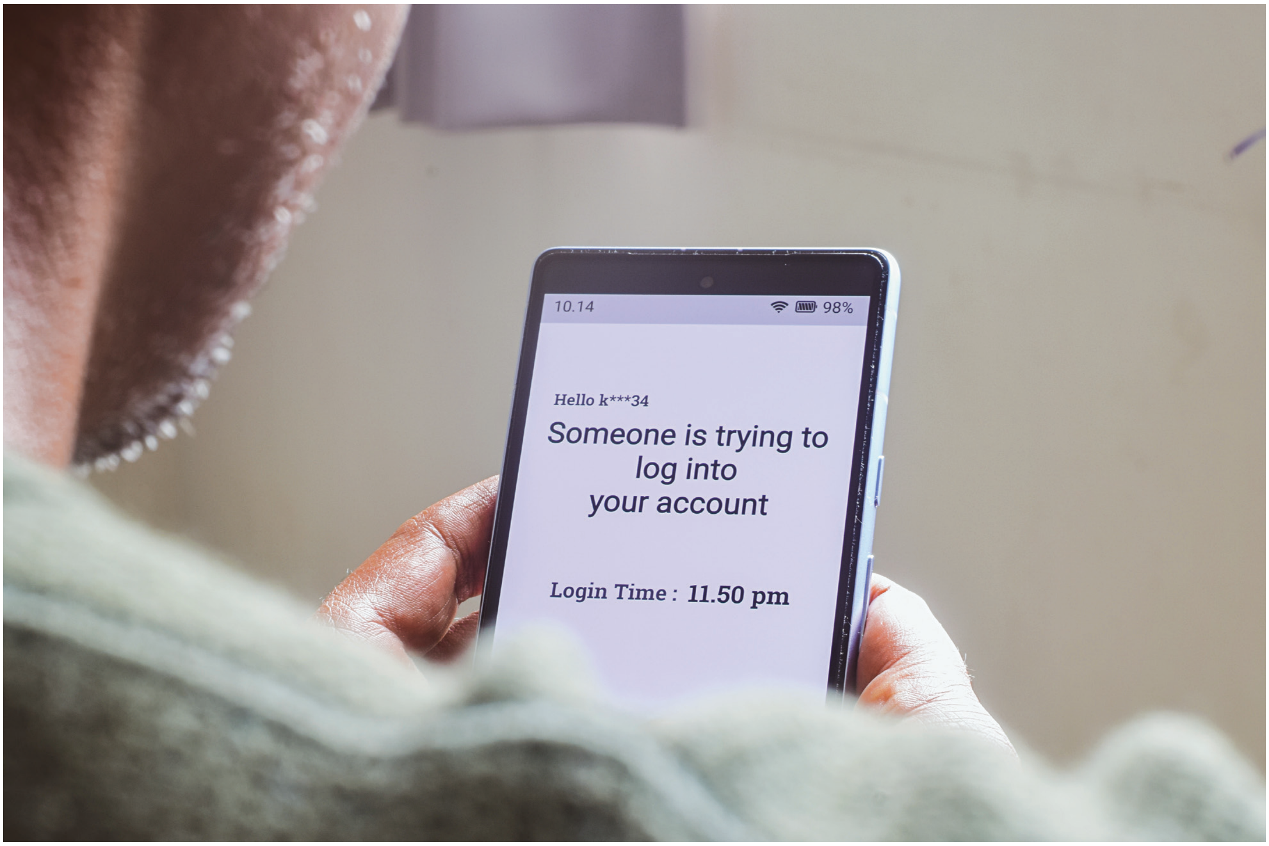

during use. Ensure graceful degradation (e.g., show an error message if camera access is denied).

3. Implement certificate pinning
- Embed server certificates in the app to block MITM attacks, as discussed in the previous chapter.

4. Conduct regular security audits
- Use static analysis (eg, MobSF) and dynamic testing tools (eg, Drozer) to identify over-privileged code paths.

5. Leverage operating-system APIs
- On iOS, utilize Data Protection classes (NSFileProtectionComplete) to encrypt sensitive files at rest.
- On Android, use EncryptedFile API from Jetpack Security to handle file-level encryption.

Monitoring and responding to app behaviour
Even well-vetted apps can become compromised; continuous monitoring helps detect anomalies.

1. Deploy Mobile Threat Defense (MTD) solutions
- Tools like Lookout, Zimperium, and Microsoft Defender for Endpoint monitor running apps and flag suspicious behavior.

2. Use firewalls with per-app rules
- On iOS, apps like Lockdown or Guardian Firewall enforce per-app network policies without root or jailbreak.
- On Android, NoRoot Firewall lets you block internet access for specific apps.

3. Review app activity logs
- On Android, check Settings → Privacy → Permission manager → Permission usage for apps accessing sensors in the background.
- On iOS, under Settings → Privacy & Security → App Privacy Report, view frequency of resource access.

Example: After installing a new flashlight app, Meera notices it attempted to access her Contacts in the background. She immediately uninstalls the app and reports it via Google Play Protect.

By rigorously vetting apps, controlling permissions at runtime, and harnessing sandboxing and containerization, you minimize the risk of unauthorized data access. Secure development practices and continuous monitoring further ensure that both first- and third-party applications remain compliant with the least-privilege principle. In the next chapter, we'll turn to network-level defenses – public Wi-Fi hardening, VPN configuration, and Bluetooth/NFC safeguards – to keep your mobile data safe in transit.